

IT IS THE CONSTANCY OF NUMBERS THAT MATTERS

ECOLOGICAL RESILIENCE

IN CONTEXT

KEY FIGURE

Crawford Stanley Holling (1930–2019)

BEFORE

1859 Charles Darwin describes the interdependence between species as an "entangled bank."

1955 In the US, Robert MacArthur proposes a measure of ecosystem stability that increases as the number of interactions between species multiplies.

1972 In contrast with MacArthur, Australian ecologist Robert May argues that more diverse communities with more complex relationships may be less able to maintain a stable balance between species.

AFTER

2003 Australian ecologist Brian Walker works with Crawford Holling to refine the definition of resilience.

The capacity for ecosystems to recover following a disturbance—such as a large fire, flood, hurricane, severe pollution, deforestation, or the introduction of an "exotic" new species—is known as ecological resilience.